Methods | MTH1002 | Term 2
Dr. Mike O'Sullivan | Year: 2025-2026

Exercise Sheet 5 (Tutorials in week 10)

Key:

- **Questions marked with a ☕** are for discussion in your tutorials.
- **Questions marked with a 🦇** are difficult questions.
- **Questions marked with a ♫** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

♫ **Exercise 1.** For each of the following first-order differential equations, check whether the given functions are solutions. Are they general solutions?

(i) $t\dfrac{\mathrm{d}x}{\mathrm{d}t} + x = t^3, \quad x(t) = \dfrac{t^3}{4} + \dfrac{A}{t}$

(ii) $\dfrac{\mathrm{d}y}{\mathrm{d}x} - y^2 = 0, \quad y(x) = Ae^x;$

(iii) $\left(\dfrac{\mathrm{d}w}{\mathrm{d}z}\right)^2 + w^2 = 2, \quad w(z) = \sin z - \cos z.$

♫ **Exercise 2.** Use the method of separation of variables to find the general solutions to the following first- order differential equations. Then also find the particular solutions that satisfy the given initial conditions.

(i) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y-1}{x-1}, \ y(2) = 3;$

(ii) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x\sqrt{1-y^2}, \ y(0) = 1.$

☕ **Exercise 3.** Find the general and particular solutions to the following linear first-order differential equations.

(i) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{y}{x} = \cos(x), \quad y(\pi/2) = 0;$

(ii) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} + 2y = x^{-4}, \quad y(1) = -\dfrac{1}{2};$

(iii) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{3}{4}y = x^2, \quad y(1) = 1.$

♫ **Exercise 4.** Find the general solutions to the following homogeneous constant-coefficient linear differential equations. For (i), also find the particular solution that satisfies the given initial condition.

(i) $y'' + 5y' + 6y = 0, \quad y(0) = 1,\ y'(0) = 1;$

(ii) $y''' - 6y'' + 11y' - 6y = 0,$ [Hint: $y = e^x$ is a solution].

♫ **Exercise 5.** For each of the following second-order constant-coefficient linear differential equations, find (a) the complementary function, (b) a particular integral, (c) the general solution, and (d) the particular solution that satisfies the given initial condition.

(i) $y'' - 7y' + 10y = x, \quad y(0) = 0,\ y'(0) = 0;$

(ii) $y'' - 3y' + 2y = 4\mathrm{e}^{3x}, \quad y(1) = 0,\ y'(1) = 0;$

(iii) $y'' - 2y' + 5y = \sin x, \quad y(0) = 0,\ y'(0) = 0.$

♫ **Exercise 6.** Solve the following partial differential equations for $u = u(x, y)$.

(i) $\dfrac{\partial u}{\partial y} = 2, \quad u(x, -1) = 1 + x^2;$

(ii) $\dfrac{\partial^2 u}{\partial x^2} = 0, \quad u(0, y) = \sin y,\ u(1, y) = 0.$

♫ **Exercise 7.** Let L be the operator

$$L[u] = 2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y^2},$$

acting on functions $u = u(x, y)$, and consider the change of variables $\xi = x + ay$, $\eta = x + by$. Find suitable parameters a, b such that the partial differential equation $L[u] = 0$ is equivalent to

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

Hence find the general solution to the PDE, giving your answer in terms of the original variables x and y.

♫ **Exercise 8.** Solve the following ordinary differential equations.

(i) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = 2\mathrm{e}^{-x}\sqrt{y},$ [Hint: Let $y = z^2$ and derive an ODE for $z(x)$];

(ii) $x\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} + x\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + \dfrac{1}{2}\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{4},$ [Hint: Let $x = t^2$, $u = e^y$, and derive an ODE for $u(t)$].